

Statistical Foundations of Prior-Data Fitted Networks [Nagler '23]

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19 June 2026

University of Freiburg

Seminar on Foundation Models for Structured Data - SS26

Group 4



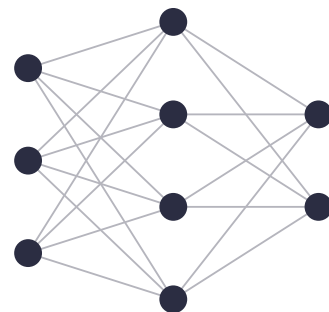


P

F

N

Network





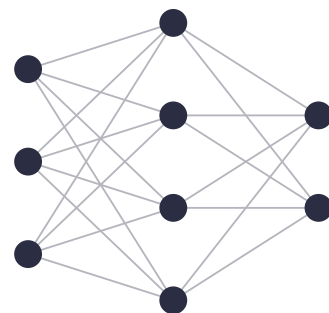
P

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Fitted

Network





P

Prior

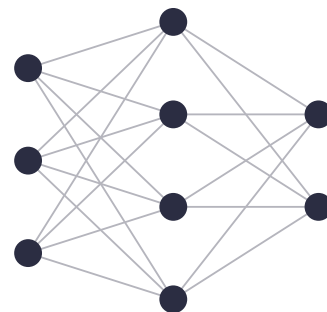
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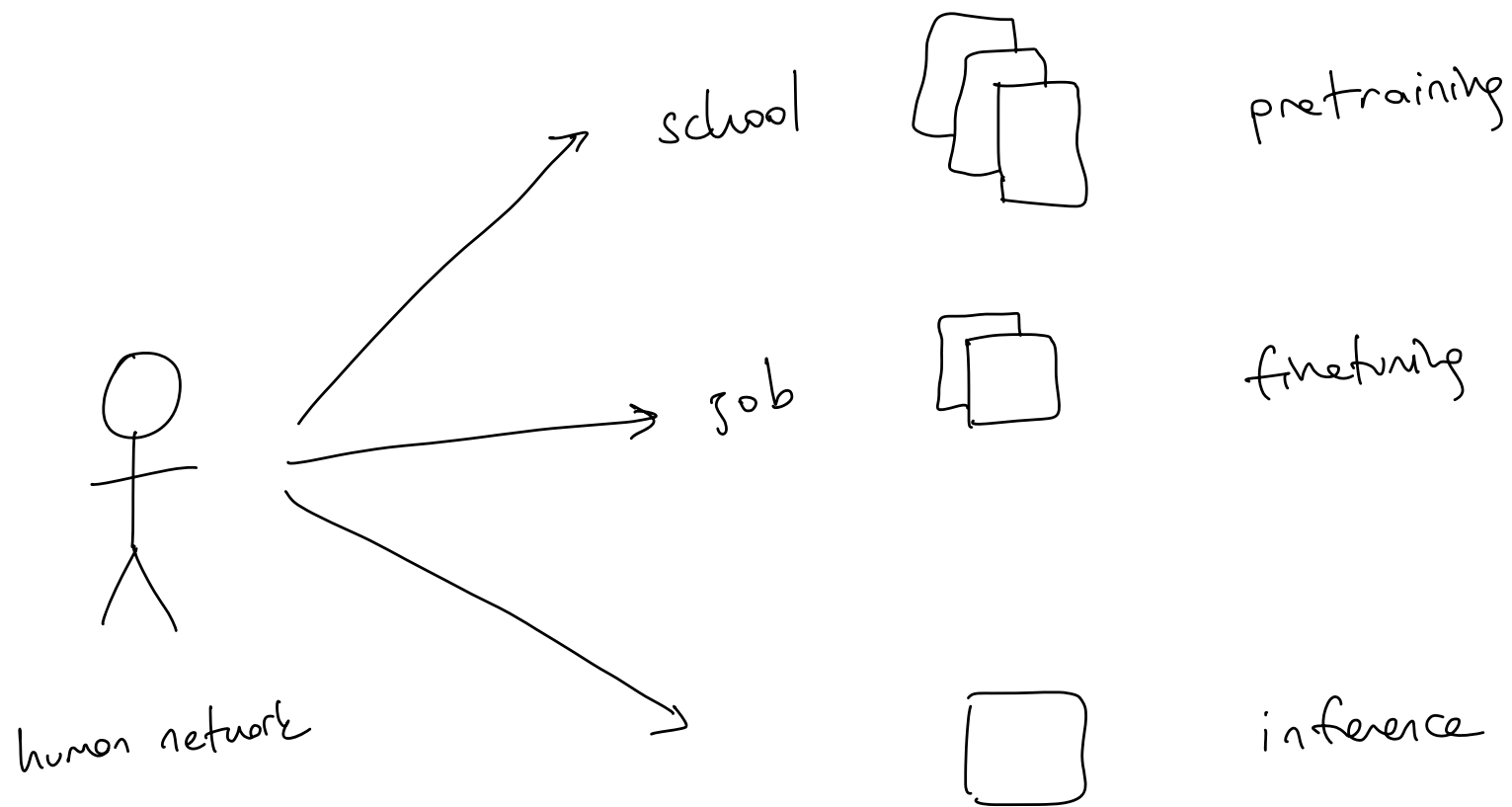
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Fitted

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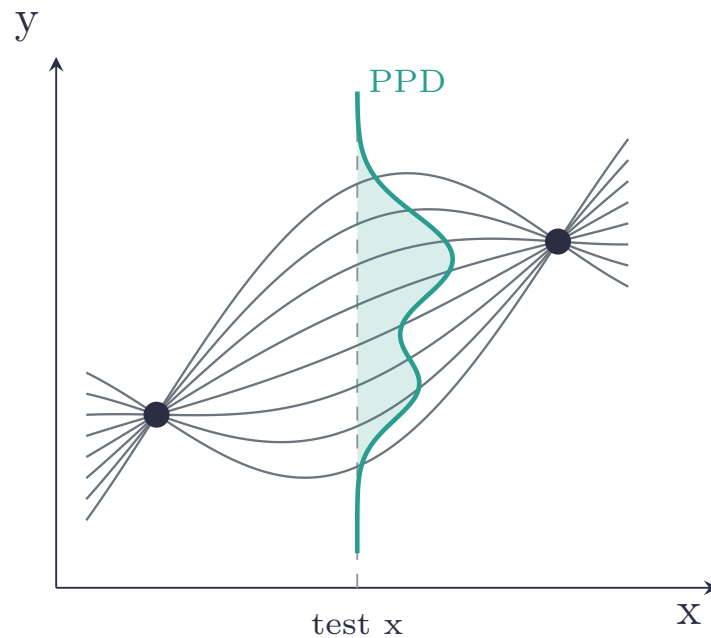
Network

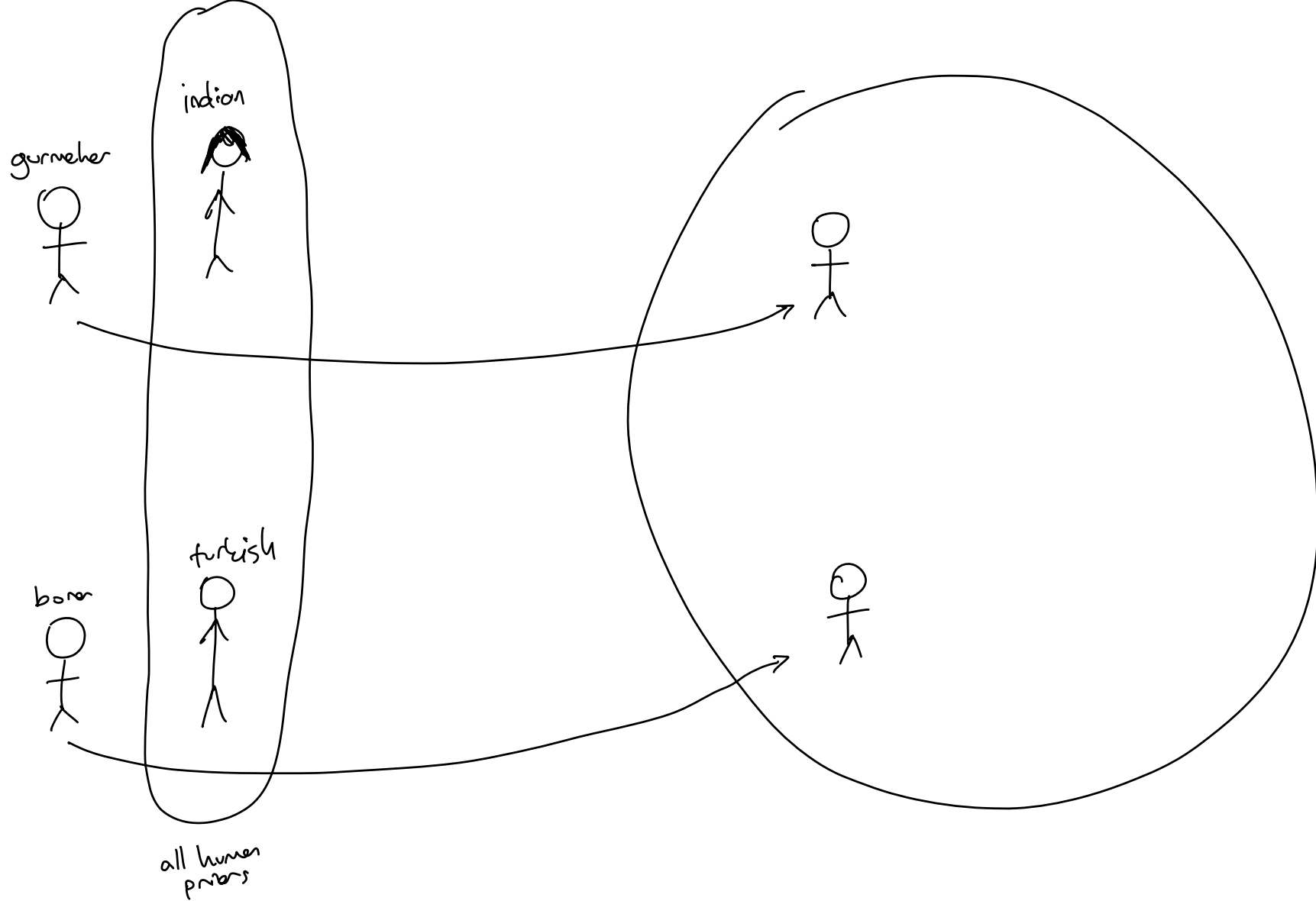




PFNs approximate PPD

- Why should you care about the PPD?
 - sum over all possible models

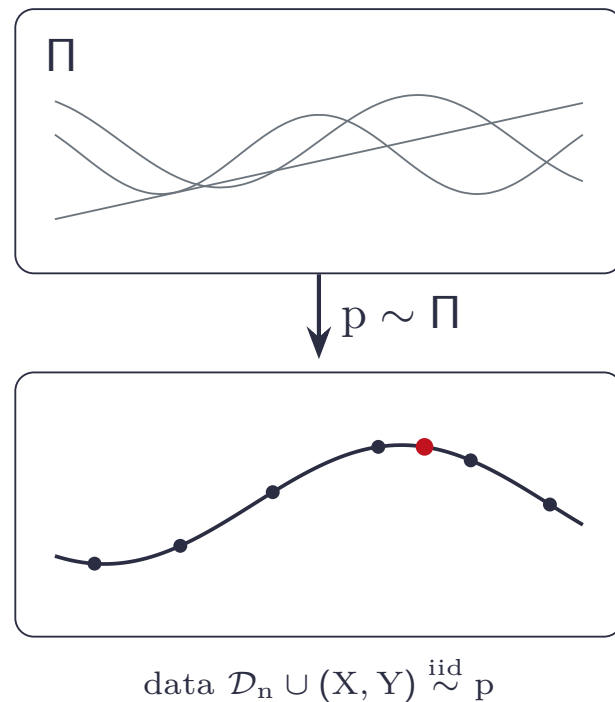




The statistical model

Data is generated in two steps:

1. draw a model $p \sim \Pi$
2. draw data $\mathcal{D}_n = (Y_i, X_i)_{i=1}^n, (Y, X) \stackrel{\text{iid}}{\sim} p$



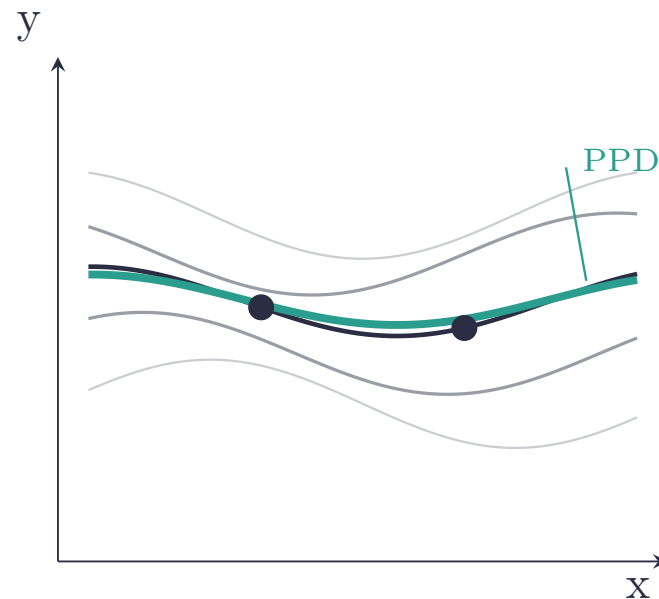
Posterior Predictive Distribution

PPD averages every model,

$$\pi(y \mid x, \mathcal{D}_n) = \int p(y \mid x) d\Pi(p \mid \mathcal{D}_n)$$

weighted by its posterior

$\Pi(p \mid \mathcal{D}_n)$: how plausible model p is, given $\mathcal{D}_n$



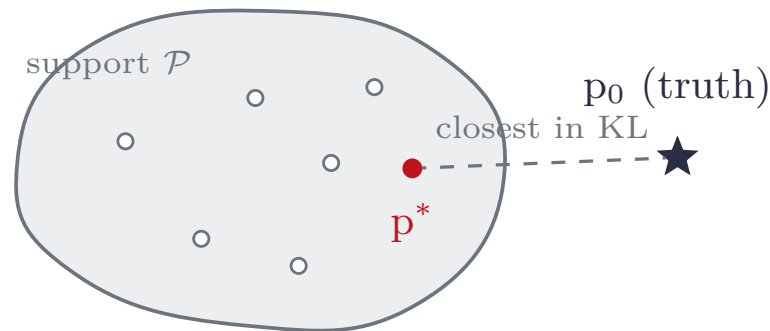
Posterior consistency

Theorem 3.1

Under (A1)–(A2), as $n \rightarrow \infty$,

$$\pi(y \mid x, \mathcal{D}_n) \longrightarrow p^*(y \mid x) \quad \text{almost surely,}$$

with p^* the **KL-optimal** model in $\mathcal{P}$.



(A1) unique $p^* \in \mathcal{P}$ minimizing $\text{KL}(p^* \parallel p_0) < \infty$

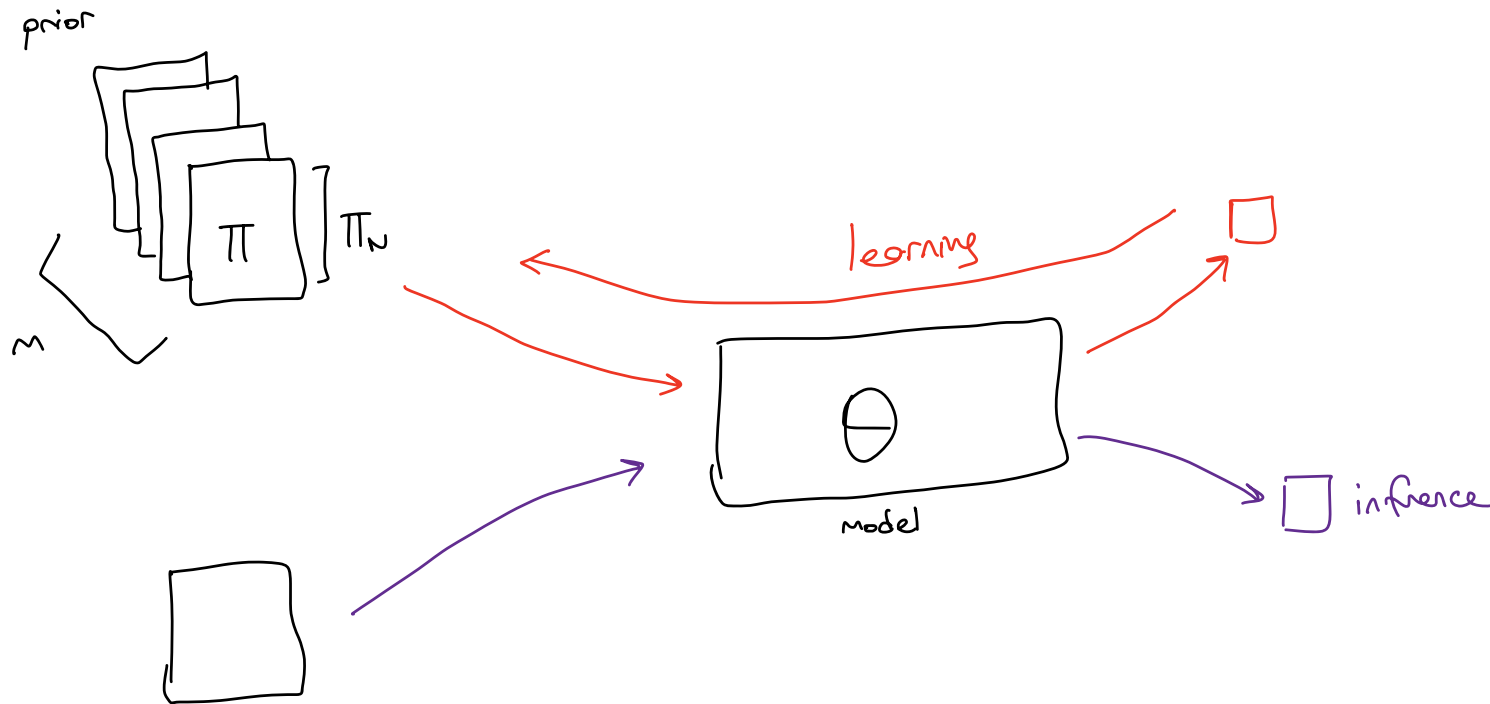
a reachable best target

(A2) $\mathcal{P}$ covered by small balls of summable prior mass

prior covers its support well

takeaway: more data $\Rightarrow$ PPD $\rightarrow$ best model the prior allows (even if $p_0 \notin \mathcal{P}$)

PFNs pretraining & inference



training size $N \sim \Pi_N$ regularizes PPD complexity (TabPFN: $N \leq 1023$)

$\hat{\theta} \rightarrow \theta^*$ as Monte-Carlo size $m \rightarrow \infty$ ($\sqrt{m}$ rate); complex models need larger m [App. B]

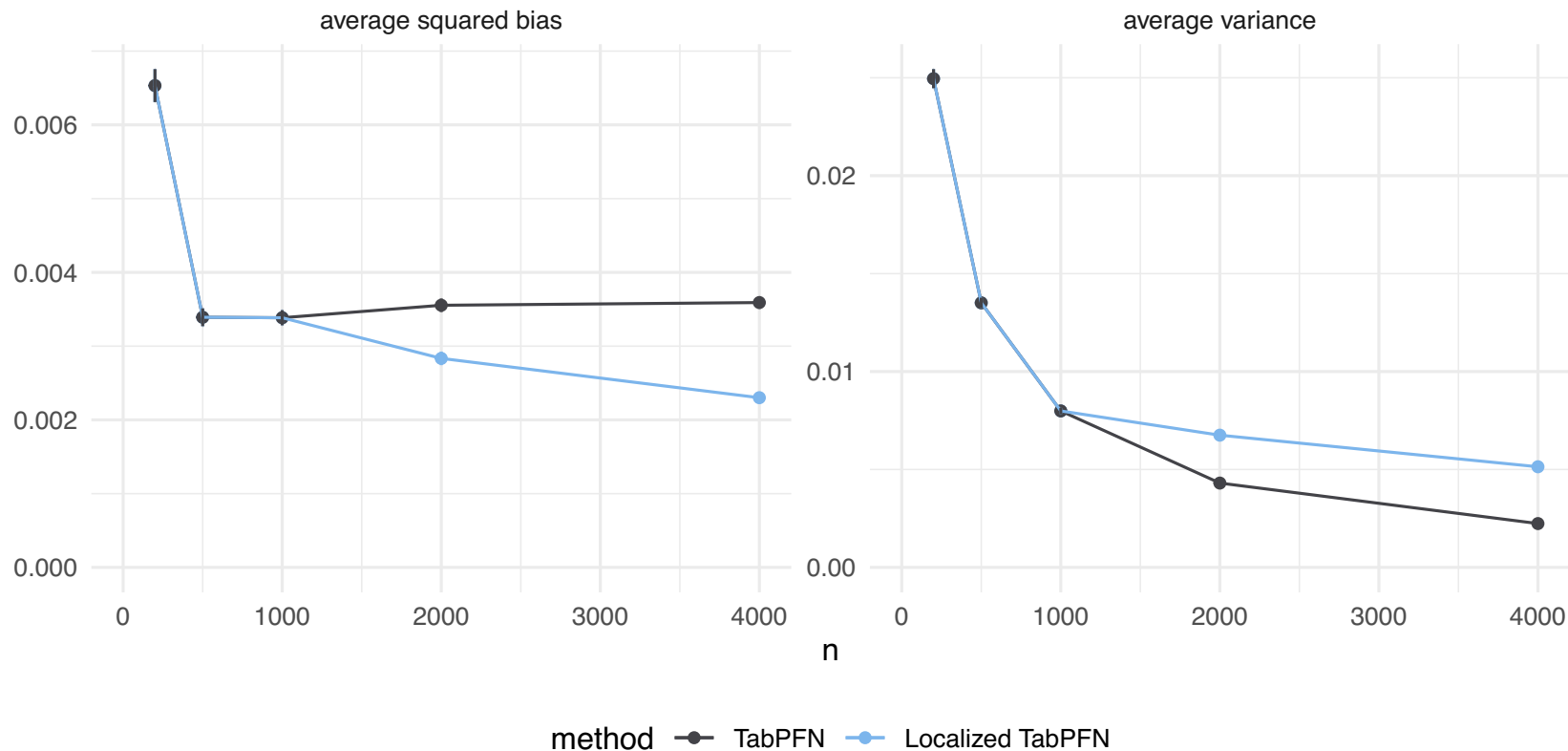
The puzzle

- no reason it behaves like a PPD for unseen sizes
- a frozen $q_{\hat{\theta}}$ at large n is not a PPD $\Rightarrow$ Theorem 3.1 no longer applies

why does a PFN pre-trained on ≤ 1000 samples
improve when fed larger datasets at inference??

- frequentist view
- bias & variance
- symmetry
- sensitivity
- locality
- other PFNs
- experiment

The paper's result [Nagler '23]

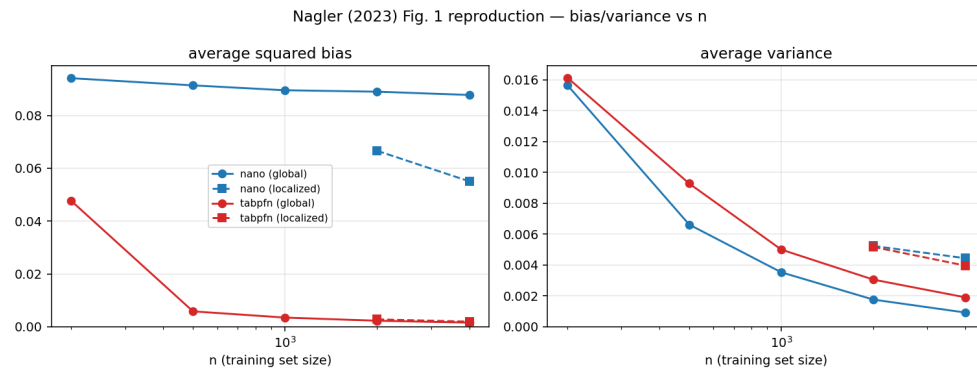


TabPFN vs localized TabPFN: variance $\sim 1/n$; bias plateaus, but localization keeps reducing it past $n = 1000$

We redid the experiment

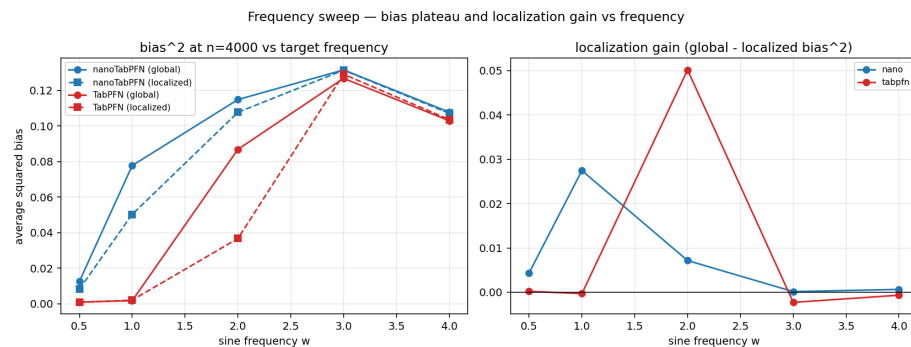
setup [Nagler '23]:

- $p_0(1 | X) = \frac{1}{2} + \frac{1}{2} \sin(1^\top X)$
- $X \sim \mathcal{N}(0, I_5)$, $Y \in \{0, 1\}$
- 100 datasets, 2 models:
 - TabPFN v2.5
 - nanoTabPFN
- avg. squared bias & variance over 100 test points

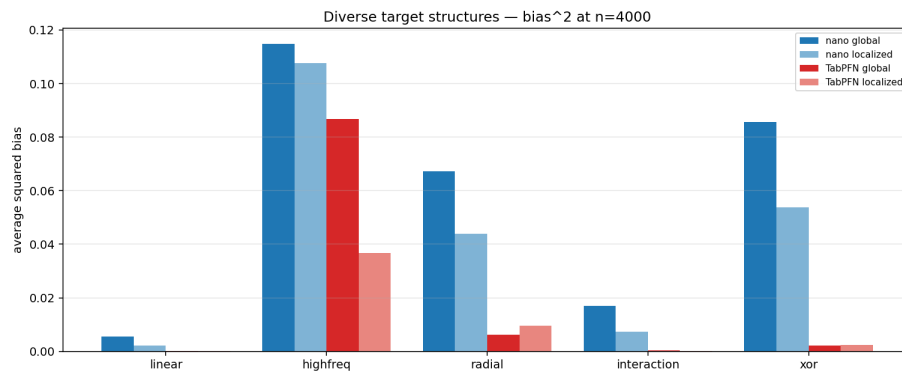


variance $\sim 1/n$; TabPFN bias drops then plateaus; localization helps the weaker model / harder targets

Beyond the paper



harder (higher-freq) targets $\Rightarrow$ bigger bias



bias & localization gain across target structures

- 2021 – PFN
- 2022 – TabPFN
- 2023 – [this paper](#)
- 2025 – TabPFNv2, nanoTabPFN, TabPFN v2.5
- 2026 – [modded nanoTabPFN](#), TabPFNv3